3. (a). Verify $\int_{-\infty}^{\infty} e^{-y^2} dy = \sqrt{\pi}$ Gaussian Integral.

$$\left(\int_{-\infty}^{\infty} e^{-y^2} dy\right)^2 = \int_{-\infty}^{\infty} e^{-x^2} dx \int_{-\infty}^{\infty} e^{-y^2} dy = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy$$

$$= \int_0^{2\pi}\int_0^{\infty} e^{-r^2} r\, dr d\theta \quad (\text{consider } e^{-(x^2+y^2)} = e^{-r^2} \text{ in polar coordinates})$$

$$= \int_0^{2\pi} d\theta \int_0^{\infty} e^{-r^2} r\, dr = 2\pi \int_0^{\infty} \frac{1}{2} e^{-r^2} dr^2 = \pi \cdot \int_{-\infty}^{0} e^s ds$$

$$= \pi \cdot e^s \Big|_{-\infty}^{0} = \pi \cdot (1-0) = \pi.$$

Therefore, $\int_{-\infty}^{\infty} e^{-y^2} dy = \sqrt{\pi}$.

(b). Verify $\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} e^{-i\lambda x} dx = e^{-i\mu\lambda - \frac{1}{2}\sigma^2\lambda^2}$.

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} e^{-i\lambda x} dx = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{(x-\mu+\sigma^2\lambda i)^2}{2\sigma^2}} \cdot e^{-\frac{1}{2}(\lambda^2\sigma^2 + 2\mu\lambda i)} dx$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-i\mu\lambda - \frac{1}{2}\sigma^2\lambda^2} \cdot \int_{-\infty}^{\infty} e^{-\frac{(x-\mu+\sigma^2\lambda i)^2}{2\sigma^2}} dx \qquad (1)$$

Gaussian Integral

$$\int_{-\infty}^{\infty} e^{-\frac{(x-\mu+\sigma^2\lambda i)^2}{2\sigma^2}} dx = \sqrt{2\sigma^2} \boxed{\int_{-\infty}^{\infty} e^{-(y+\alpha i)^2} dy} \quad \text{where } \begin{cases} y = \frac{x-\mu}{\sqrt{2}\sigma} \\ \alpha = \frac{\sigma\lambda}{\sqrt{2}} \end{cases} \qquad (2)$$

Let $I = \int_{-\infty}^{\infty} e^{-(y+\alpha i)^2} dy$, $I_L = \int_{-L}^{L} e^{-(y+\alpha i)^2} dy$. (3)

$$I_L = \int_{-L}^{L} e^{-(y+\alpha i)^2} dy = \int_{-L-\alpha i}^{L-\alpha i} e^{-z^2} dz = \int_{-L-\alpha i}^{L-\alpha i} e^{-z^2} dz - \int_{-L}^{L} e^{-z^2} dz + \int_{-L}^{L} e^{-z^2} dz$$

Let $\hat{I}_L = \int_{-L}^{L} e^{-z^2} dz$

$$I_L - \hat{I}_L = \int_{-L-\alpha i}^{L-\alpha i} e^{-z^2} dz - \int_{-L}^{L} e^{-z^2} dz.$$

Consider a complex contour $C: -L \xrightarrow{①} L \xrightarrow{②} (L-2i) \xrightarrow{③} (-L-2i) \xrightarrow{④} (-L)$.

$$\oint_C = \int_{-L}^{L} + \int_{L}^{L-2i} + \int_{L-2i}^{-L-2i} + \int_{-L-2i}^{-L} = \int_{-L}^{L} - \int_{-L-2i}^{L-2i} - \int_{L-2i}^{L} + \int_{-L-2i}^{-L}$$

$$I_L - \hat{I}_L = \int_{-L-2i}^{L-2i} - \int_{-L}^{L} = -\oint_C + \int_{L-2i}^{L} - \int_{-L-2i}^{-L} = \int_{L-2i}^{L} - \int_{-L-2i}^{-L}$$

By 3(a). $\lim_{L\to\infty} \hat{I}_L = \lim_{L\to\infty} \int_{-L}^{L} e^{-y^2} dy = \int_{-\infty}^{\infty} e^{-y^2} dy = \sqrt{\pi}$

$$\lim_{L\to\infty} (I_L - \hat{I}_L) = \lim_{L\to\infty} \left(\int_{L-2i}^{L} e^{-z^2} dz - \int_{-L-2i}^{-L} e^{-z^2} dz \right) = 0 - 0 = 0$$

The above limit is easily seen to vanish.

So, $\lim_{L\to\infty} I_L = \lim_{L\to\infty} \hat{I}_L = \sqrt{\pi}$ (3)

By (1) (2) (3), we prove

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} e^{-i\lambda x} dx = e^{-i\lambda\mu - \frac{1}{2}\sigma^2\lambda^2}$$

(C). Find the Fourier transformation of $e^{-\eta x^2}$, for any constant $\eta > 0$.

$f(x) = e^{-\eta x^2}$.

$$\hat{f}(k) = \int_{-\infty}^{\infty} e^{-\eta x^2} \cdot e^{-ikx} dx = \int_{-\infty}^{\infty} e^{-(\eta x^2 + ikx)} dx$$

$$= \int_{-\infty}^{\infty} e^{-\left(\sqrt{\eta}x + \frac{ik}{2\sqrt{\eta}}\right)^2} \cdot e^{\left(\frac{ik}{2\sqrt{\eta}}\right)^2} dx = e^{-\frac{k^2}{4\eta}} \int_{-\infty}^{\infty} e^{-\left(\sqrt{\eta}x + \frac{ik}{2\sqrt{\eta}}\right)^2} dx$$

$$= e^{-\frac{k^2}{4\eta}} \cdot \int_{-\infty}^{\infty} e^{-(\sqrt{\eta}x)^2} dx = \frac{1}{\sqrt{\eta}} e^{-\frac{k^2}{4\eta}} \int_{-\infty}^{\infty} e^{-z^2} dz$$

$$= \sqrt{\frac{\pi}{\eta}} \cdot e^{-\frac{k^2}{4\eta}}$$

(d). $\begin{cases} u_t = a^2 u_{xx} \\ u(x,0) = \phi(x) = u^0(x) \end{cases}$

$g_t(x) := N(x; 0, 2a^2 t)$ is heat kernel.

Show $u(x,t) = (g_t * \phi)(x)$ is solution.

$\partial_t u(x,t) = a^2 \partial_x^2 u(x,t)$

$\partial_t \hat{u}(k,t) = a^2 \partial_x^2 \hat{u}(k,t) = a^2 \cdot (ik)^2 \hat{u}(k,t) = -a^2 k^2 \hat{u}(k,t)$ since $\hat{f'}(k) = ik \hat{f}(k)$

$\begin{cases} \frac{d}{dt} \hat{u}(k,t) = -a^2 k^2 \hat{u}(k,t) \\ u(x,0) = \phi(x) = u^0(x) \end{cases} \Rightarrow \hat{u}(k,t) = e^{-a^2 k^2 t} \cdot \hat{u}(k,0) = \hat{g}_t(k) \cdot \hat{u}(k,0)$

where $\hat{g}_t(k) = e^{-a^2 k^2 t}$.

Since $\widehat{(f * g)}(k) = \hat{f}(k) \cdot \hat{g}(k)$

$u(x,t) = (g_t * u^0)(x) = g_t(x) * \phi(x)$. where $g_t(x)$ is heat kernel.

Since $\hat{g}_t(k) = e^{-a^2 k^2 t}$.

$g_t(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{g}_t(k) e^{ikx} dk = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(a^2 t \cdot k^2 - ix \cdot k)} dk$

$= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(a\sqrt{t}k - \frac{ix}{2a\sqrt{t}})^2 + (\frac{-ix}{2a\sqrt{t}})^2} dk = \frac{e^{-\frac{x^2}{4a^2 t}}}{2\pi} \int_{-\infty}^{\infty} e^{-(a\sqrt{t} \cdot y)^2} dy$

$= \frac{e^{-\frac{x^2}{4a^2 t}}}{2\pi} \cdot \frac{1}{a\sqrt{t}} \int_{-\infty}^{\infty} e^{-z^2} dz = \frac{e^{-\frac{x^2}{4a^2 t}}}{2a\pi\sqrt{t}} \cdot \sqrt{\pi} = \frac{1}{2a\sqrt{\pi t}} e^{-\frac{x^2}{4a^2 t}}$

Therefore,

$u(x,t) = g_t(x) * \phi(x) = \int_{-\infty}^{\infty} \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} \cdot \phi(\xi) \, d\xi$

$= \frac{1}{2a\sqrt{\pi t}} \int_{-\infty}^{\infty} \phi(\xi) e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi$

$f * g = \int_{-\infty}^{\infty} f(x-\xi) g(\xi) d\xi$